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A systematic review of foresight methods used in the formulation of futures scenarios: evidence from articles indexed in scopus and web of science (2014–2023)

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Abstract

This study presents a systematic literature review of the foresight methods used in the formulation of futures scenarios, analyzing 115 articles indexed in Scopus and Web of Science between 2014 and 2023. Applying the PRISMA methodology, the review identifies a plural methodological ecosystem in which—although qualitative approaches predominate—there are patterns of combination toward methodological triangulation aimed at strengthening plausibility and reducing bias in complex environments. The analysis shows that the most frequently used methods are environmental scanning (PESTEL), participatory workshops, the Delphi method, structural analysis (MICMAC), and Causal Layered Analysis (CLA). A significant expansion of these practices is observed beyond the traditional corporate sphere into sectors such as public health, higher education, governance, and environmental sustainability. The study further highlights an incipient presence of methods that incorporate artificial intelligence, text mining, and computational models, which complement qualitative narratives with quantitative rigor. The article proposes the concept of methodological aggregation capacity to describe the ability of scenario methodologies to coherently incorporate techniques of different epistemic natures. In conclusion, the maturity of the field rests on the systemic convergence between data-driven rigor and narrative depth, consolidating scenarios as instruments for social learning and strategic decision-making.

Keywords Futures scenarios, Systematic review, Methodological triangulation, Methodological aggregation capacity, Strategic foresight

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Introduction

Futures scenarios definitions have evolved from essentially exploratory formulations, focused on describing possible or plausible futures in highly uncertain environments [8, 26, 82], toward integrative approaches that combine exploration with the formulation of preferred futures and strategic action [28, 35, 42, 43].

At the outset, futures scenarios were conceived as alternative narratives that allowed practitioners to visualize the consequences of different trends and potential discontinuities [14, 56, 79]. Later, researchers and practitioners incorporated normative and participatory approaches, transforming scenario studies into spaces for co-creation, social learning, and deliberation about preferred futures. Simultaneously, quantitative and semiquantitative methods were articulated into these exercises, broadening the methodological scope of the scenarios and adding rigor to the construction of plausible futures [25, 30, 61, 71].

Futures scenario exercises have thus transited from foresight diagnostic and exploratory tools into participatory, scientifically grounded methodological devices that integrate research, innovation, and decision-making in contexts of high complexity and uncertainty. Within these exercises, qualitative, quantitative, and semiquantitative methods converge to capture both the subjective and objective dimensions of the systems under study. This convergence has generated methodological diversity, placing the various schools of scenario building in dialogue. First, the structural approaches. The French School relies on structural, actor-strategy, and morphological analyses, conducted using a suite of software tools such as MICMAC, MACTOR, and SMIC-PRO. Second, qualitative approaches. The Intuitive Logics School focuses on qualitative methods for the participatory, narrative construction of alternative futures. Third, integrated contemporary approaches. These approaches integrate Delphi, simulation, modeling, and multi-criteria evaluation, among others, to articulate statements about possible futures.

The methodological aggregation capacity demonstrated in futures scenario exercises refers to the coherent functional articulation of qualitative methods with either quantitative or semiquantitative methods. This feature constitutes both the principal challenge and the greatest strength of these methodologies. Although practitioners increasingly combine new methods, a preference for trend-based futures persists, which weakens the capacity to anticipate disruptive or discontinuous futures.

The formulation of futures scenarios has become a pivotal tool for strategic planning in highly uncertain environments [5]. For scenarios to fulfill this purpose, their statements must be coherent, consistent, and plausible [13, 15, 28, 59].

This systematic literature review aims to analyze the patterns of foresight methods used in futures scenario exercises across different processes, sectors, and thematic domains, drawing on academic articles indexed in Scopus and Web of Science (WoS) between 2014 and 2023, with particular attention to how qualitative, quantitative, and semiquantitative methods have been combined. To this end, the analysis examines: (1) the foresight methods most frequently used in scenario formulation; (2) the sectors in which scenario exercises are concentrated; and (3) the patterns of methodological articulation identified in the literature.

Conceptual framework

Futures scenarios are methodological instruments that allow systematic exploration of possible and plausible futures and, in some approaches, preferred futures, to support strategic decision-making in the present [6, 11]. Scenarios differ from forecasting, which seeks a single most probable future, and from extrapolation, which projects historical trends, because they deliberately engage with the plurality and uncertainty of the future.

The field is organized around several central approaches: (1) Intuitive Logics, which privileges qualitative narrative and the participation of key stakeholders [18]; (2) the French School, which combines structural, actor-strategy, and morphological analyses under a voluntarist conception of the future [33]; (3) critical and transformative approaches, which articulate Causal Layered Analysis (CLA), the Futures Triangle, and the six pillars of critical foresight to deconstruct implicit assumptions and explore alternative futures (source); and 4) Alternative Futures [24], which defines four possible futures to facilitate a strategic foresight process for identifying, designing, and moving toward a preferred future. Since the first decade of the twenty-first century, these approaches have been complemented by computational, quantitative, and implementation-oriented approaches (backcasting, road mapping, system dynamics, and text mining) that expand the analytical capacity of scenario exercises without compromising the interpretive dimension that is constitutive of the field.

Methodological aggregation capacity

This article introduces the concept of *methodological aggregation capacity*, which is defined as the ability of scenario formulation to incorporate techniques of diverse nature in a coherent and functionally integrated manner, while maintaining internal consistency and plausible results. Methodological aggregation capacity represents an intermediate level of coupling, situated between the coexistence of independent techniques and full integration into a single formal framework. In contrast to triangulation, which seeks validation through convergence,

aggregation distributes complementary functions among techniques organized around an anchor method, thereby structurally enriching scenario construction.

Epistemological position of the study

The epistemological position adopted in this review is pluralist and non-hierarchical with respect to the identified methodological approaches. The review does not assume that one approach is inherently superior to another; rather, each responds to specific purposes, contexts, and value systems. The evaluation criterion is the internal coherence among the exercise's objectives, the selected methods, and the results obtained.

Methodology

Methodological approach

This systematic literature review examines foresight methods used in futures scenario exercises published in academic articles indexed in Scopus and WoS between 2014 and 2023, as well as the sectors and thematic domains in which these exercises have been applied and the ways in which qualitative and quantitative methods have been combined. The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol, which is widely used for this type of article in the social and applied sciences.

Information sources

The search was conducted across two high-impact, internationally recognized academic databases: Scopus and Web of Science (WoS). These databases were selected for their multidisciplinary coverage and for their focus on publications in the highest-ranked quartiles (Q1 and Q2), which guarantees the scientific quality of the exercises reviewed.

Search strategy

The search strategy centers on the keywords foresight, future, and scenarios. Using Boolean operators, the following search string was constructed:

Foresight and scenarios and future

The search was limited to texts published between 2014 and 2023, classified in Scopus as articles and conference papers, and in WoS as articles and review articles.

Inclusion and exclusion criteria

The inclusion criteria were as follows:

- Articles published between 2014 and 2023.
- Articles that explicitly applied a methodology for the formulation of futures scenarios with associated foresight methods.
- Publications indexed in Scopus and WoS.
- Documents reporting minimum information on the application sector, the foresight method used, the number of futures scenarios formulated, and time horizons.

Articles meeting any of the following conditions were excluded:

- Articles falling outside the defined temporal range.
- Papers not directly related to the formulation of futures scenarios.
- Theoretical or conceptual papers, without presenting a methodology for scenario formulation.
- Duplicate publications in both Scopus and WoS.

Selection process

The four phases of the PRISMA methodology [58] were followed: (1) Identification: 610 articles were retrieved from Scopus and 710 from WoS, for a total of 1,320 articles. (2) Screening: 409 duplicate texts were identified using the Mendeley reference manager, leaving 911 papers, of which 796 were excluded after reading titles and abstracts. (3) Eligibility: 115 articles were reviewed and systematized in an Excel matrix (see Fig. 1).

Data extraction and systematization

Data extraction was conducted in two stages: initially through a review of the authors, followed by the use of NotebookLM to identify patterns within the texts. The extracted information was organized using a foresight methods extraction matrix, which included the following variables: year of publication, article title, sector or thematic area, foresight methods employed, number of scenarios formulated, and time horizon. These variables served as inputs for subsequent analysis.

Methodological combination patterns

Combination patterns were generated through a co-occurrence analysis of the methods used in each article and recorded in the data extraction matrix. After examination, these patterns were classified into seven exhaustive and mutually exclusive categories, each named according to the foresight school or logic characteristic of the constituent methods. If an article combined methods from more than one pattern (25.2%), it was assigned to the family that structured variable selection and scenario construction.

Data analysis

The analysis was developed in two stages: first, descriptive quantitative analysis, which calculated absolute and relative frequencies of methods, sectors, and methodological combinations; and second, exploratory and comparative analysis, which examined the co-occurrence of

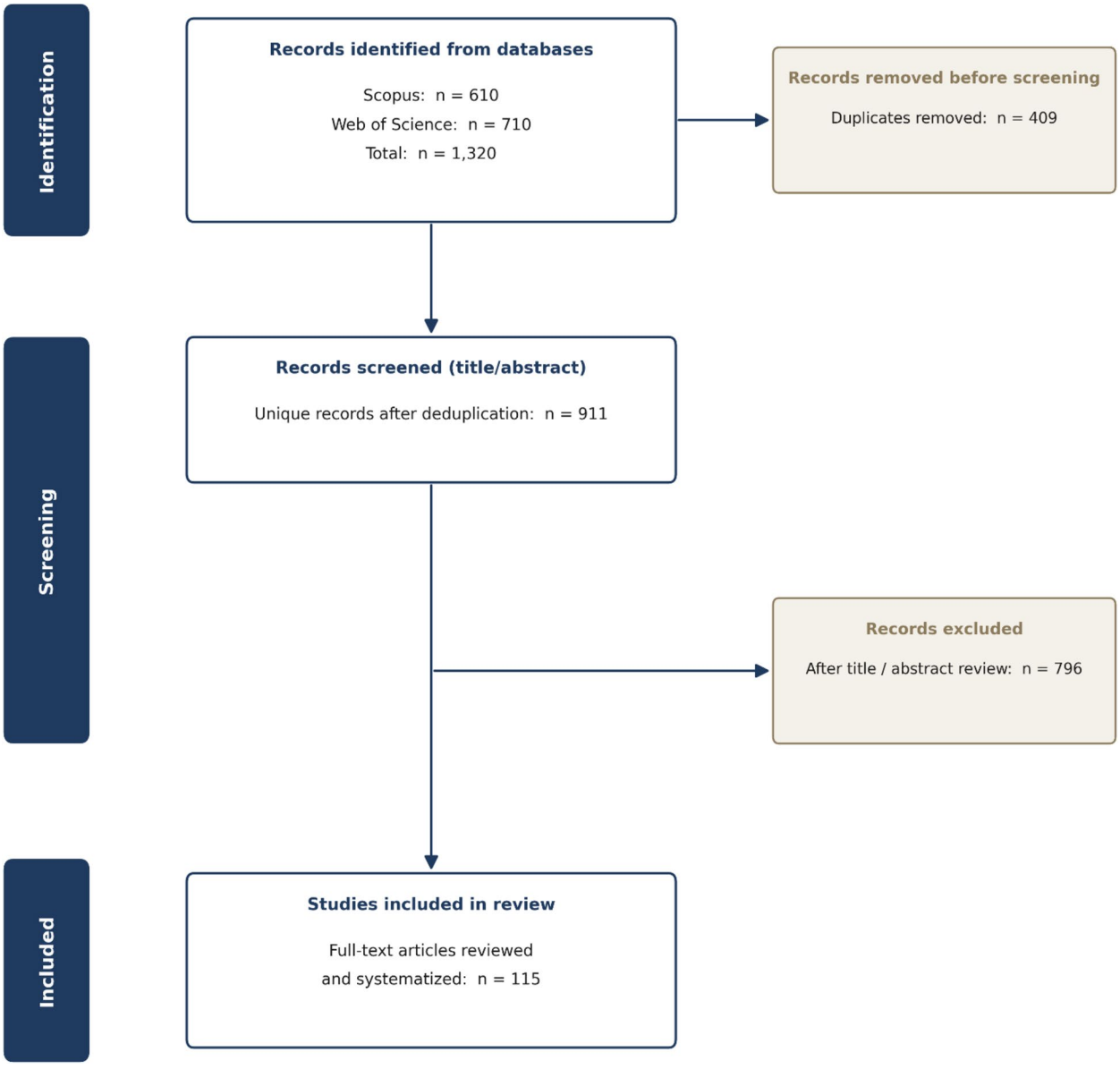


Fig. 1 PRISMA flow diagram. Source: Authors

sectors and methods, the co-occurrence of quantitative and qualitative methods, and the co-occurrence and transversal distributions during the period under study.

Corpus size and adequacy

After analyzing the 115 articles in blocks of 10, we found that, starting with article 100, the number of new sectors begins to decrease, allowing us to conclude that this corpus is sufficient to cover their diversity (Table 1).

Saturation is observed across sectors; the number of new sectors begins to decline starting with Article 100, allowing us to conclude that the 115 articles cover the field's sectoral diversity. There is no evidence of saturation regarding methods, as their number continues to increase, and the analyzed corpus presents a unique combination of 115 methods: a result partly due to the fact that method names were not standardized in the data extraction matrix.

Limitations

Regarding the methods analyzed: as the number of texts increased, so did the number of new methods, partly because the names of these methods and their combinations were not standardized; hence, each of the 115 articles has a unique composition.

Table 1 Accumulation of methods, sectors, and methodological combinations

Cumulative articles	Cumulative unique methods	Increase in methods	Cumulative unique sectors	Increase in sectors	Unique methodological combinations
10	29	—	10	—	10
20	55	+26	19	+9	20
30	76	+21	27	+8	30
40	98	+22	33	+6	40
50	119	+21	41	+8	50
60	140	+21	50	+9	60
70	165	+25	57	+7	70
80	184	+19	65	+8	80
90	205	+21	74	+9	90
100	232	+27	81	+7	100
110	264	+32	86	+5	110
115	274	+10*	88	+2*	115

Compiled by the authors based on the foresight methods extraction matrix described in the methodology

Results

The systematic review of the 115 selected articles identified clear patterns across the period under study regarding the most frequently used foresight methods, sectors of application, and patterns of methodological combination among qualitative, quantitative, and semiquantitative methods.

Foresight methods most frequently used in the formulation of futures scenarios

Futures scenarios do not seek to predict the future; rather, they aim to establish a dialogue among key stakeholders to develop a shared frame of reference for possible futures [38]. In the analyzed articles, the 15 most frequently used methods were identified and grouped into two blocks based on frequency.

Foresight methods most frequently used in the formulation of futures scenarios

In the analyzed articles, the 15 most frequently used methods were identified and grouped into two blocks based on frequency: methods applied in articles with a frequency of 5% or higher were classified as dominant (Block A), and methods based on computational or AI techniques were classified as emerging (Block B) (Table 2).

Block a: dominant methods

Four of these methods, used in 49% of the analyzed exercises, illustrate the dominance of qualitative approaches and the participatory orientation of the methodologies used to formulate futures scenarios (see Fig. 2 below) (Table 3).

Table 2 The eight most frequently used foresight methods

Method	Type	Role in the scenario exercise	Representative sources
Environmental scanning (PESTEL/STEEPV)	Qualitative	Identifies driving forces in the macro-environment and key variables for scenario formulation	Alizadeh et al. [5]; Van Vuuren [77]; Pasi & Misuraca [60]
Participatory workshops	Qualitative	Information gathering and results validation through knowledge exchange and collective deliberation	Kamprath & Mietzner [45]; Shantiko et al. [68]
Delphi method	Semiquantitative	Iterative expert consensus to identify strategic variables and validate trends	Gary & von der Gracht [29]; Schüll [69]; McGrath & Fischetti [52]
Structural analysis (MICMAC)	Semiquantitative	Identifies key system variables by their influence and dependence; sets the foundation for scenario formulation	Alizadeh et al. [5]; Del Rio Cortina et al. [27]; Marín-González et al. [51]
Causal Layered Analysis (CLA)	Qualitative	Deconstructs reality into data, systems, worldviews, and myths; champions transformative futures	Inayatullah & Song [40]; Inayatullah [42]; Milojević & Inayatullah [53]
Morphological analysis	Semiquantitative	Unpacks the system into variables to explore possible future combinations	Nair, Kai Wen, & Mei Ling [56]; Johansen [44]; Bootz et al. [14]
Quantitative modeling and simulation	Quantitative	Uses mathematical models to simulate scenarios and estimate probabilities under different hypotheses	Aparicio, Urbano, & Gómez [7]; Hafezi et al. [36]
Expert panels and key informants	Qualitative	Allows capturing multidisciplinary and tacit knowledge	Bock & Bontoux [13, 72]

Compiled by the authors from the foresight methods extraction matrix described in the methodology

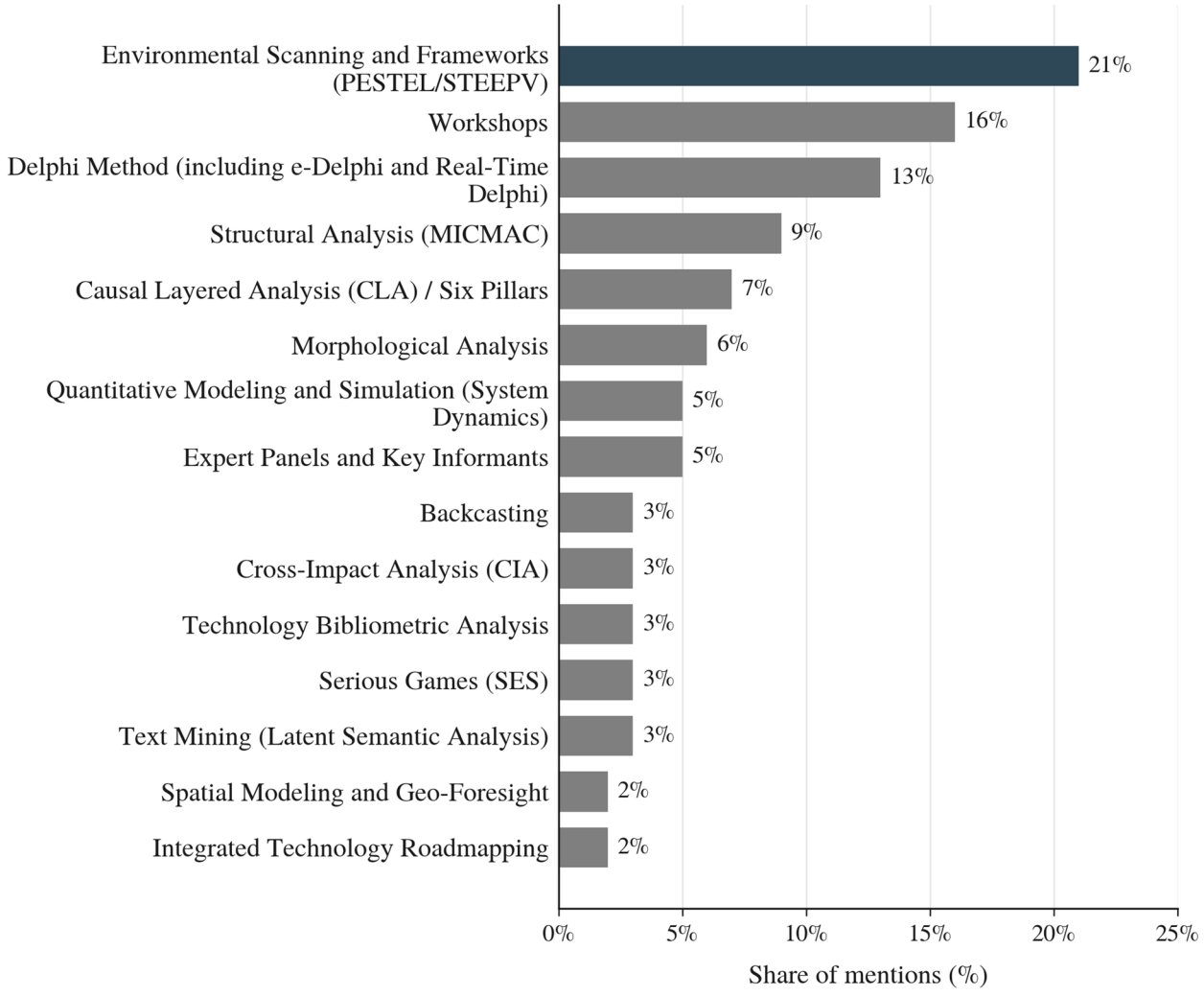


Fig. 2 Foresight Methods Most Frequently Used in the Formulation of Futures Scenarios (2014–2023). Note. Compiled by the authors from the foresight methods extraction matrix described in the methodology

Block B: emerging methods supported by advanced technologies

The inclusion of these methods in futures scenario exercises evidences the systematic incorporation, that is, the methodological aggregation, of new technologies such as serious games, spatial modeling, bibliometric analysis, and data mining, which enable researchers to process large volumes of information and identify weak signals that experts might otherwise overlook.

Taken together, the fifteen methods identified illustrate the methodological aggregation capacity that characterizes futures scenario exercises: the ability to coherently integrate techniques of a qualitative nature with quantitative ones or with emerging methods supported by artificial intelligence, thereby creating a plural methodological ecosystem that adapts to new technologies without losing its internal coherence or its strategic purpose. And the coherent functional articulation of qualitative techniques with quantitative or emerging artificial intelligence-based ones creates a plural methodological ecosystem that incorporates new technologies without losing its internal coherence or strategic purpose.

Sectors and processes in which futures scenario exercises are formulated

The range of applications for futures scenario exercises has increased during the period under study: scenarios are no longer monopolized by large corporations and the economic sphere. They have migrated into social, academic, and territorial sectors, thereby strengthening strategic planning and institutional sustainability. Approximately 70% of the exercises are concentrated in one cross-cutting process and six sectors (Table 4):

The remaining 30% of scenario exercises is distributed across additional sectors, confirming that futures scenarios are used not only to address large-scale

Table 3 Seven additional methods supported by emerging technologies

Method	Type	Role in the scenario exercise	Representative sources
Backcasting	Qualitative–normative	Examines the past to understand trajectories and to formulate preferred futures by working backward from a desired end state	Boschetti, Fulton,& Grigg [12]; Mohammed et al. [54]; Nash et al. [57]
Cross-impact analysis	Quantitative	Assesses the probabilistic interdependence among variables to anticipate possible future effects	Vollmar et al. [79]; Taheri Demneh, Zackery, & Nouraei [75]
Bibliometric analysis	Quantitative	Analyzes scientific production through metrics on publications, citations, and research trends	Stelzer et al. [71]; Vishnevskiy et al. [78]; Barbosa et al. [8]
Serious games	Experiential	Simulates possible futures through structured games to support collaborative strategic decisions	Bontoux et al. [17]; Westphal et al. [81]
Text mining	Quantitative–computational	Identifies patterns and foresight-relevant knowledge from large volumes of textual data	Kwon, Kim, & Park [47]; Snijders et al. [70]; Kayser & Shala [46]
Computational and Spatially Explicit Foresight	Quantitative–spatial	Projects future geographic scenarios through the analysis of territorial variables	Low [50]; Lestrelin et al. [49]
Integrated technology roadmapping	Mixed	Connects anticipation with implementation through goal-oriented technology roadmaps	Hussain, Tapinos, & Knight [38]; Gerdsri et al. [30]

Compiled by the authors from the foresight methods extraction matrix described in the methodology

Table 4 Sectors and processes accounting for 70% of futures scenario exercises

Sector/Process	Approx. %	Representative sources
Innovation (cross-cutting process)	18%	Christofilopoulos & Mantzanakis [21]; Munigala, Oinonen,& Ekman [55]
Public health	14%	Vollmar et al. [79]; Gauthier & Wac [31, 35, 37, 80]; Cruz-Aguilar & Medina-Vásquez [22]
Governance	12%	Inayatullah & Song [40]; Bellamy & Healey [10]; Pasi & Misuraca [60]
Environment	8%	Graveline et al. [34]; Ibáñez Blancas et al. [39]; Bauwens, Hekkert,& Kirchherr [9]
Higher education	8%	Inayatullah & Milojević [41]; Schüll [69]; Stolze & Sailer [72]
Energy transition	7%	Proskuryakova & Ermolenko [63]; Tatar, Kalvet, & Tiits [74]; Alizadeh et al. [5]
Agriculture	7%	Pilorgé& Muel [61]; Bourgeois et al. [15]; Ajilore& Fatunbi [3] ; Camara, Bourgeois, & Jahel [20]; Alfonso Carvajal & Garrido Rubiano [4]

Compiled by the authors from the foresight methods extraction matrix described in the methodology

Table 5 Sectors accounting for the remaining 30% of futures scenario exercises

Sector/Process	Approx. %	Representative sources
Urban mobility	5%	Sessa & Ricci [67]; Burrieza-Galán et al. [19]; Rogers & Hunt [65]
Agro-industry	4%	Pilorgé & Muel [61]; Ginanneschi [32]
Climate change	4%	Bontoux & Bengtsson [16]; Stratigea & Katsoni [73]; Low [50]; Labboux, Sales, & Valette [48, 66]; Daniels, Grim, & Morgan [23]
Tourism	4%	Riensche et al. [64]; Tuominen & Ascencao [76]
Forestry	4%	Packalen, Kärkkäinen, & Toppinen [59]; Shantiko et al. [68]
Territorial development	4%	Abdoli, Habib, & Babazadeh [1]; Acosta et al. [2]; Lestrelin et al. [49]; Del Rio Cortina et al. [27]; Shantiko et al. [68]
Basic education	3%	Snijders et al. [70]; McGrath & Fischetti [52]
Shipbuilding industry	2%	Vishnevskiy et al. [78]; Dekhtyaruk et al. [26]

Compiled by the authors from the foresight methods extraction matrix described in the methodology

systemic transitions but also as tools for optimizing decisions in operational and territorial contexts, where uncertainty manifests in more focused yet equally critical ways (Table 5).

Methodological combination patterns

Text analysis allowed us to identify seven patterns of methodological combinations based on co-occurrence (how often various methods appear together in the same article), which are presented in order from highest to lowest frequency of occurrence in the corpus. Each pattern was given a name based on the logic that links the methods comprising it.

Pattern 1: intuitive logics

This is the most frequently used combination of methods: literature review, environmental scanning (in its PESTEL and STEEPV variants), identification of drivers of change, the 2 × 2 matrix, and scenario storylines. This pattern reaffirms the dominance of the Anglophone school in

this type of exercise, which is distinguished by its application of qualitative methods and by its understanding of scenarios not as predictions but as coherent stories about how the future may unfold.

Pattern 2: french school

This pattern combines morphological analysis, actor-strategy analysis, and the Régnier Abacus, all of which are associated with the French School founded by Gaston Berger and consolidated by Michel Godet. The corresponding software toolkit—MICMAC for structural analysis, MACTOR for actor-strategy analysis, and Morphol for morphological analysis, among others—helps reduce subjectivity in strategic decision-making about the future.

Pattern 3: expert consultation with quantitative triangulation

This pattern combines the Delphi method (in its various modalities), expert panels, participatory panels, and focus groups. Although Delphi is typically qualitative in its consultation phase, it is often triangulated with statistical analyses (medians, quartiles, and percentages) to assess relationships among variables and with documentary or bibliometric analyses to validate identified trends. This triangulation enhances the validity of the results.

Pattern 4: multi-level critical foresight

This pattern combines methods, including the Six Pillars, Causal Layered Analysis (CLA), and the Futures Triangle [42, 53]. Its multi-level approach allows researchers not only to describe foresight methodologies but also to interpret them critically, transcending linear views of the future and fostering a complex, systemic understanding of it. Its strength lies in deconstructing metaphors and worldviews that implicitly orient decision-making.

Pattern 5: implementation-oriented (backcasting)

This pattern comprises methods such as backcasting, technology roadmapping, flex points, and operational scenarios. The pattern responds to the need to connect anticipation with action, moving from an exploration-oriented to a transformation-oriented formulation of scenarios. It thus offers a robust methodology for constructing preferred futures with explicit roadmaps.

Pattern 6: computational and spatially explicit foresight

It comprises bibliometrics, text mining, system dynamics, Monte Carlo simulation, and spatial modeling. It is situated on the 'evidence' side of Popper's [62] diamond and groups methods that apply algorithmic procedures to massive textual, numerical, or georeferenced data, formalizing the system under study in an explicit representation (corpus and semantic field, causal structure of stocks and flows, probability distributions, or layers and territorial units), from which descriptions of its state or conditional projections of its behavior are obtained. This concept differs from the literature review defined in Pattern 1.

Pattern 7: participatory, experiential, and co-creative

This pattern comprises serious games, LEGO Serious Play, card games, and landscape simulations. These methods are oriented to exploring futures through interaction, play, and the collective construction of meaning, thereby facilitating social learning, the internalization of tacit knowledge, the articulation of shared visions of the future, and the exploration of possible futures from diverse perspectives. They contribute to the legitimacy and social ownership of scenario exercises by actively involving multiple stakeholders in their construction.

Co-occurrence patterns in the use of methods for designing futures scenarios confirms that, although methodologies based solely on qualitative methods remain widely used, there is a clear pattern toward triangulation and methodological aggregation. This pattern increases the validity, explanatory power, and transformative potential of futures studies. It also reflects methodological maturity in the field, where different approaches coexist not as competitors but often tend to be combined to triangulate sources of information and biases, to plausibly formulate possible futures, and contribute to strategic decision-making in contexts of uncertainty (Fig. 3).

Integration of methods by their epistemic nature: qualitative, quantitative, and semi-quantitative

Contemporary methodologies for the formulation of futures scenarios are grounded in an amalgam of qualitative, quantitative, and semiquantitative methods approaches, with methodological triangulation serving as a recognized benchmark for quality. Exercises relying exclusively on qualitative methods remain prevalent, particularly when the aim is to shift social perspectives and deconstruct/unpack underlying metaphors. Quantitative and semiquantitative methods, in turn, add objectivity to these narratives.

A salient finding is the absence of studies that employ exclusively quantitative methodologies in the corpus. This gap suggests an implicit consensus among scenario practitioners: while quantitative data may secure technical accuracy, it gains strategic footing only when embedded within a coherent narrative framework that renders it socially meaningful and actionable for contemporary decision-making. Accordingly, the soundness of scenario methodologies lies in the systematic convergence between analytical rigor and narrative depth (see Fig. 4).

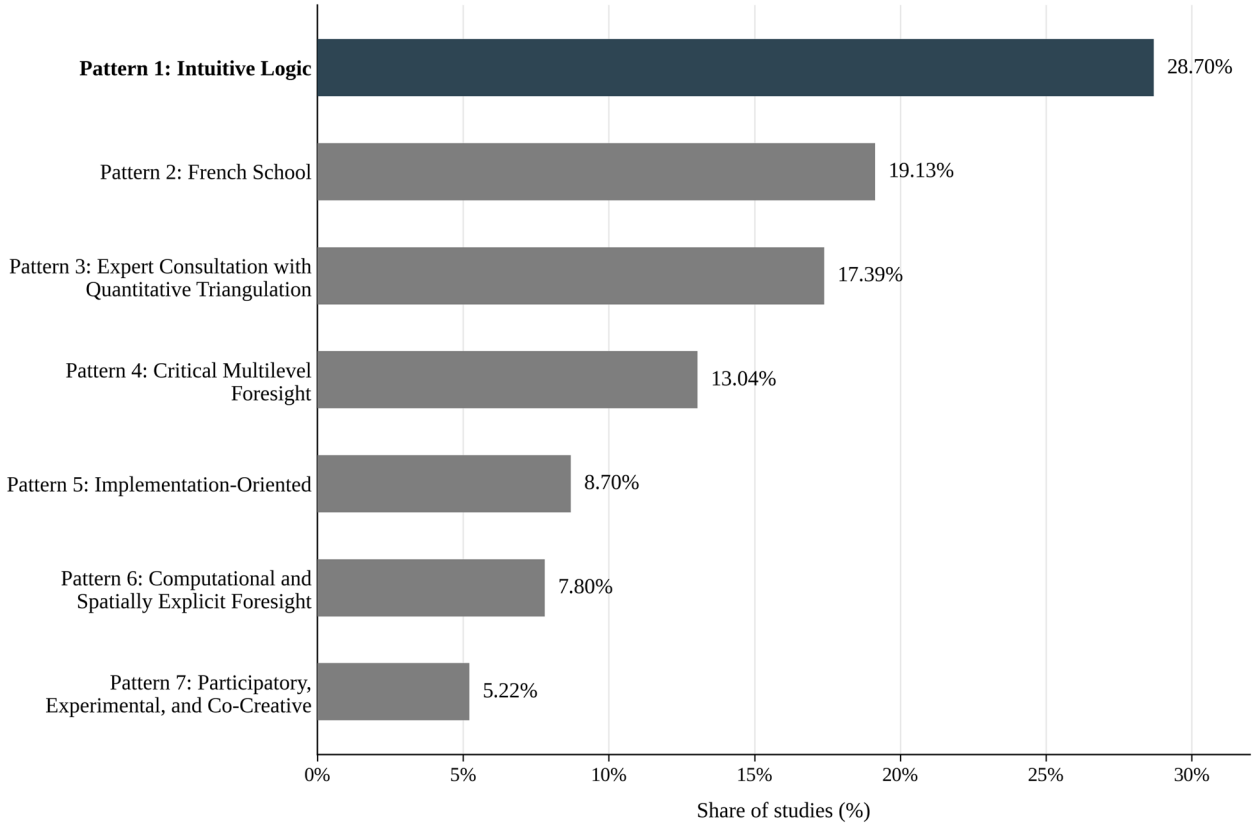


Fig. 3 Most Frequently Used Combinations of Methods in the Formulation of Futures Scenarios (2014–2023). Note. Compiled by the authors from the foresight methods extraction matrix described in the methodology

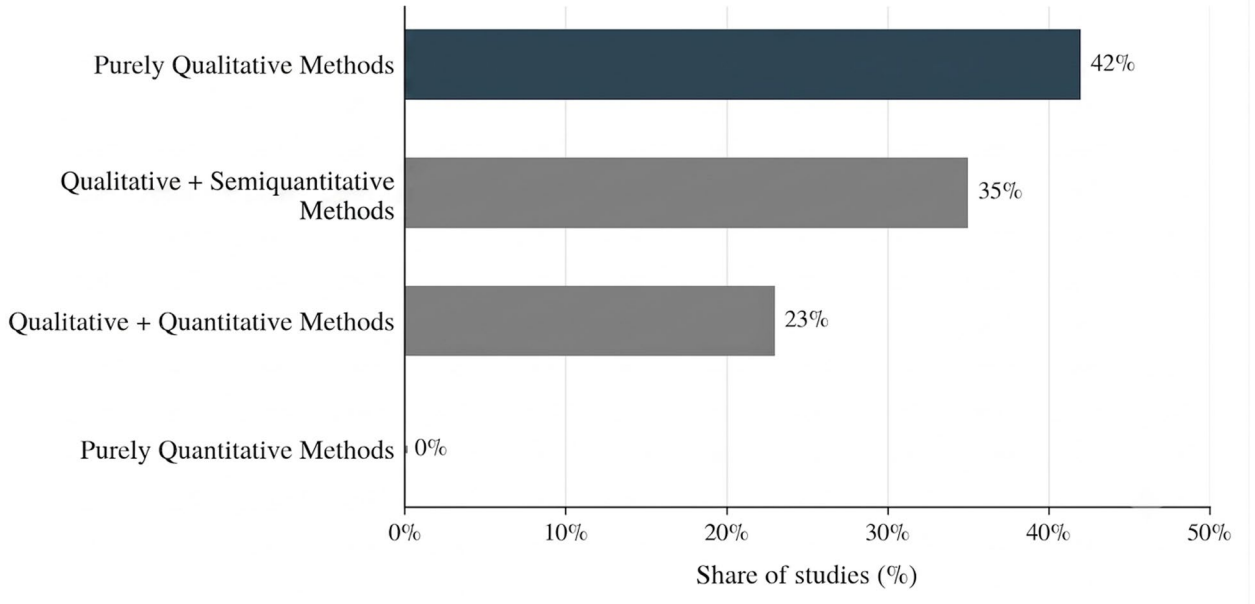


Fig. 4 Combination of Qualitative, Quantitative, and Mixed Methods Used in the Formulation of Futures Scenarios (2014–2023). Note. Compiled by the authors from the foresight methods extraction matrix described in the methodology

Table 6 Artificial Intelligence the temporal boundary of the corpus

Indicator	Findings from the articles reviewed
Articles incorporating AI or computational components	88 articles (76.5%)
Most frequently used AI methods	Text mining, spatial modeling, and automated bibliometrics
Period of highest concentration	2017–2023
Sectors with the highest AI adoption	Technology, health, and national/regional foresight
Predominant function of AI within the methodologies	Complementary, not substitutive, of expert judgment

Compiled by the authors from the foresight methods extraction matrix described in the methodology

Artificial Intelligence as the temporal boundary of the corpus

Although the aim of this article is not to analyze artificial intelligence (AI) across foresight methodologies, of the 115 articles, 88 (76.5%) were published between 2014 and 2021, while 97 (84.3%) were published before 2023. This indicates an incipient and still limited presence of techniques incorporating some form of AI or computational component (Table 6).

Integrating AI into scenario methodologies will enable the analysis and synthesis of large volumes of textual and numerical data that would be unmanageable for human analysts. Such processing capacity will help surface weak signals that experts might otherwise overlook due to cognitive biases or procedural limitations. Nevertheless, the exercises analyzed indicate that this integration is conceived as complementary rather than as a substitute for human judgment, making clear that AI improves the quality of inputs but does not replace or displace collective deliberation or the narrative construction of possible futures.

Discussion

The findings of this systematic review warrant further discussion of four salient dimensions: dominance of qualitative methods, methodological aggregation capacity, sectoral expansion, and the incorporation of emerging technologies.

Dominance of qualitative methods and its epistemological foundations

The reviewed methodologies show a clear dominance of qualitative methods, a finding consistent with the prior reviews of Bishop, Hines, and Collins [11] and of Amer, Daim, and Jetter [6] which document the global preeminence of the Intuitive Logics School. This dominance should not be interpreted as a limitation but as an epistemological choice. Futures scenarios are, at their core, social construction instruments, and qualitative methods are best suited to capturing the subjective dimensions—perceptions, values, and worldviews—that guide strategic decision-making.

The distinctive contribution of this study, compared to prior reviews, is not merely its confirmation of the predominance of qualitative methods, but rather its demonstration of the coherent integration of qualitative approaches with quantitative and semiquantitative methods. This articulation produces what we have termed *methodological aggregation*: qualitative methods are not displaced by quantitative ones; rather, there is a functional articulation to yield higher-quality results.

Methodological aggregation capacity as a structural feature of the field

The identification of seven combination patterns, ranging from Intuitive Logics to computational and co-creative approaches, shows that the field is not fragmented into irreconcilable schools but rather operates as a structured combinatorial repertoire. The seven patterns coexist and are usually combined; each responds to specific purposes, contexts, and actors with different capabilities.

This conclusion complements and updates the findings of Amer, Daim, and Jetter [6] who identified a growing methodological hybridization in the field. The contribution of the present study is to show that such hybridization is neither eclectic nor random, but rather structured into recognizable patterns. This structuring provides a clearer basis for methodological selection in relation to the aims, context, and stakeholders involved in each scenario exercise.

Sectoral expansion and its implications

The expansion of futures scenario exercises from the corporate sphere into sectors such as education, health, governance, and the environment, as documented in this review, suggests that futures scenarios are becoming a public methodological tool accessible to all social actors, communities, and public and private organizations. This finding corroborates the observation made by Amer, Daim, and Jetter [6] that scenario planning has expanded into fields beyond the military and corporate sectors, which were once regarded as its exclusive domains.

This expansion, however, spawns a capabilities challenge: not all sectors that adopt scenarios possess technical teams sufficiently competent to ensure the coherence, consistency, and plausibility of the resulting statements. This finding reinforces the importance of foresight training programs and of technical cooperation between academic institutions and other organizations.

Artificial intelligence: opportunity and challenge for the field

The identification of methods that incorporate AI anticipates a deeper transformation of the field as large

language models (LLMs) and other forms of Generative AI become more widespread. This transformation should not be interpreted as a threat to the deliberative and participatory dimensions of scenarios; rather, it should be understood as an expansion of the information-processing capacities that feed them.

The finding that no article in the corpus adopts a purely quantitative methodology suggests that the community of practice in futures scenarios has already implicitly established a principle of complementarity: AI improves inputs but does not replace human judgment. This principle should be made explicit and formalized as AI tools are increasingly incorporated into scenario exercises.

Foresight research agenda and theoretical practice (2025–2036)

Although computational methods remain marginal (7.8%) and have not demonstrated a sustained upward trajectory, we observe that the emerging skill in foresight is transitioning from manual technique application to the design and critical curation of AI-assisted analyses. In this context, expert judgment serves as the certifying authority for assessing the plausibility and theoretical relevance of results.

Limitations

The search equation prioritized English-language outputs containing the term "foresight," which may have led to the underrepresentation of Latin American and Francophone contributions. Regarding the methods analyzed, the corpus was reviewed in blocks of ten articles. As the number of texts increased, the diversity of identified methods also grew. This increase was facilitated by the lack of standardization in method names and combinations, resulting in each of the 115 articles presenting a unique methodological composition.

Conclusions

The qualitative approach predominates, complemented by the cross-sectional application of quantitative and semi-quantitative methods. This methodological triangulation serves as a standard of convergence, articulating subjective and objective dimensions to generate more plausible, consistent, and coherent scenarios.

This systematic review identifies a methodological ecosystem characterized by flexibility and adaptability, which we have conceptualized as *methodological aggregation capacity*. Within this ecosystem, seven patterns of methodological combination coexist: Intuitive Logics, the French School, expert consultation, multi-level critical foresight, implementation-oriented backcasting, quantitative–computational foresight, and participatory–experiential approaches. Far from competing, these patterns complement one another, forming a structured combinatorial repertoire that enables the discipline to remain at the knowledge frontier by absorbing new tools.

The application of futures scenarios has transcended traditional domains, penetrating sectors such as health, education, environment, governance, agriculture, and territorial development. This diversification illustrates that scenarios have become a tool for addressing complex problems across multiple contexts, especially those characterized by high uncertainty and a need for long-term planning.

Of the 115 exercises, 48, representing 42%, used only qualitative methods; 40, equivalent to 35%, used quantitative and semi-quantitative methods; and 27 of them, equivalent to 23%, incorporated both qualitative and quantitative methods in their work, and none of the exercises used only quantitative techniques. What is noteworthy is not the predominance of the qualitative, but rather that, despite the existence of a wide range of computational tools, most of the exercises still do without them.

The incorporation of artificial intelligence into futures scenario exercises represents the principal opportunity for the methodology's future development. Its incorporation should be conceived as complementary to—not substitutive of—human judgment, maintaining the balance between analytical rigor and interpretive depth. Researchers in the field are invited to further explore this incorporation in future work.

Declaration of AI-assisted technologies in the writing process

The authors used Grammarly, an AI-assisted writing tool, solely to improve the manuscript's grammar, spelling, and rhetorical clarity, and NotebookLM to support the identification of thematic patterns across documents during data extraction (see Data Extraction and Systematization). All NotebookLM outputs were independently verified by the authors against the source articles before being included in the data extraction matrix. NotebookLM was not used to generate conclusions, references, or the interpretive constructs proposed in this article (for example, methodological aggregation capacity), which reflect the authors' own analysis.

Author Contribution

The author(s) read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

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